

The Petroleum Industry

Petroleum is the primary source of energy in the world today. In addition to this, to hydroelectric power, Natural gas, Nuclear power and other sources.

Petroleum is also a source of numerous organic chemicals used to manufacture drugs, Fertilizers, Clothing and many other products or chemicals - popularly called Petrochemicals. Petroleum and Coal are the two main sources of hydrocarbons.

Petroleum is an refined viscous dark-brown liquid often called Crude oil. It is a complex mixture of alkanes, alkenes, cycloalkanes and aromatic compounds. Petroleum was formed in the Earth's crust over the course of millions of years by anaerobic decomposition of animal and vegetable matter by bacteria.

Petroleum deposits is widely distributed throughout the world; however, they are found mostly in the middle East, North America, Russia, China, Mexico, Brazil, Venezuela and of course in some countries in Africa like Nigeria, Libya, Sudan, Algeria, Chad, Niger, etc.

The actual composition of petroleum varies with location. Pennsylvania crude oils in the USA are mostly aliphatic hydrocarbons, major components of western crude oils are aromatic in nature.

Nigerian crude is naphthenic, thus very light and with very low sulphur content. Crude oil was first discovered in commercial quantity in 1956 by Shell in Orléansville-Bağda state. After decades of exploration by the Nigerian Bitumen Company and Shell ~~Diary~~ ^{Diary},

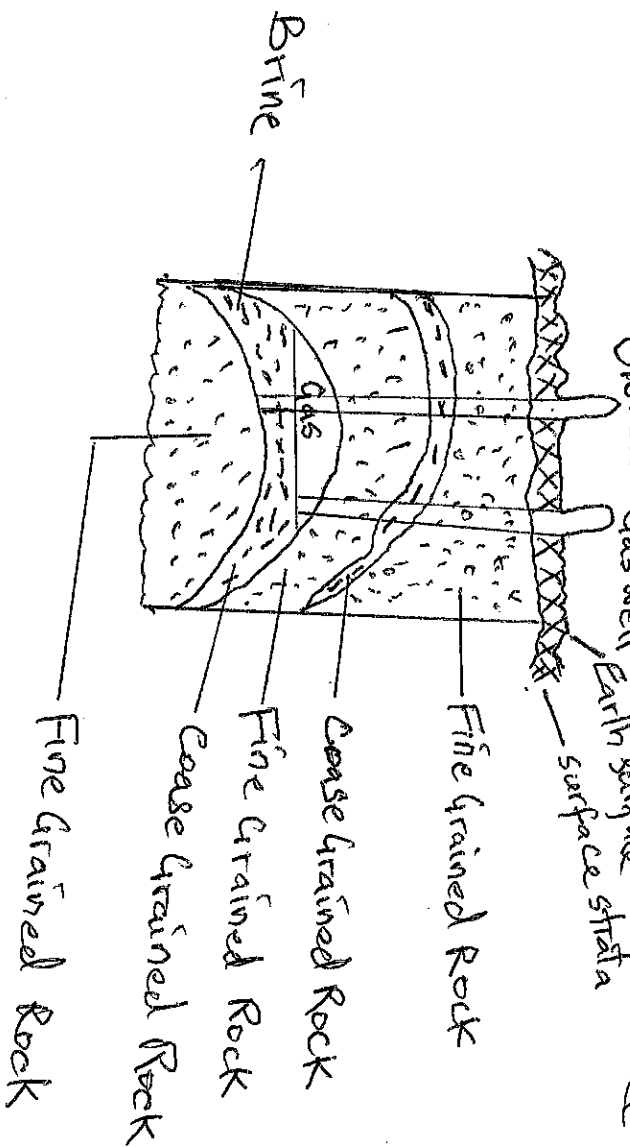
Petroleum components can be classified according to the range of their boiling points even though it literally contains thousands of hydrocarbon compounds.

These hydrocarbons can be separated on the basis of molecular mass by fractional distillation.

Heating crude oil to about 400°C converts the viscous oil to hot vapour and liquid fluid. In this form it enters the fractionating tower. The vapour rises and condenses on various collecting trays according to the temperatures at which the various components of the vapour liquefy. Some gases are drawn off at the top of the column, and the unvaporized residual oil is collected at the bottom as residue which can be further fractionated to obtain more refined products, ^{Petrol or gasoline} is probably the best known Petroleum product. It is a mixture of volatile hydrocarbons and contains mostly alkanes, cycloalkanes and a few aromatic hydrocarbons. Some of these compounds are far more suitable for fuelling an automobile engine than others. Thus we need for further treatment and refinement of gasoline or petrol.

Generally, Petroleum is trapped in impervious rocks at depths from 100 to 3,500 meters and often floats on salt water. The oil is brought to the surface by drilling in the earth's crust and simultaneously fixing pipes until the drill penetrates the oil bearing sands. Sometimes the petroleum rushes out through them due to high pressure.

Together with the oil a certain amount of gas is generally associated in the form of a gas cap. The gas is generally known as Natural Gas. See diagram;



As seen as the pressure of the Petroleum subsides, air pressure is applied through pumps to force the oil out of the well.

The above process is called drilling and subsequent extraction of crude from the oil well or fields. From the oil fields, the crude Petroleum is sent to oil refineries through pipelines having diameter of 200 meters underground.

The depths of petroleum deposits vary from place to place. The first commercial oil well, known as Drake well,

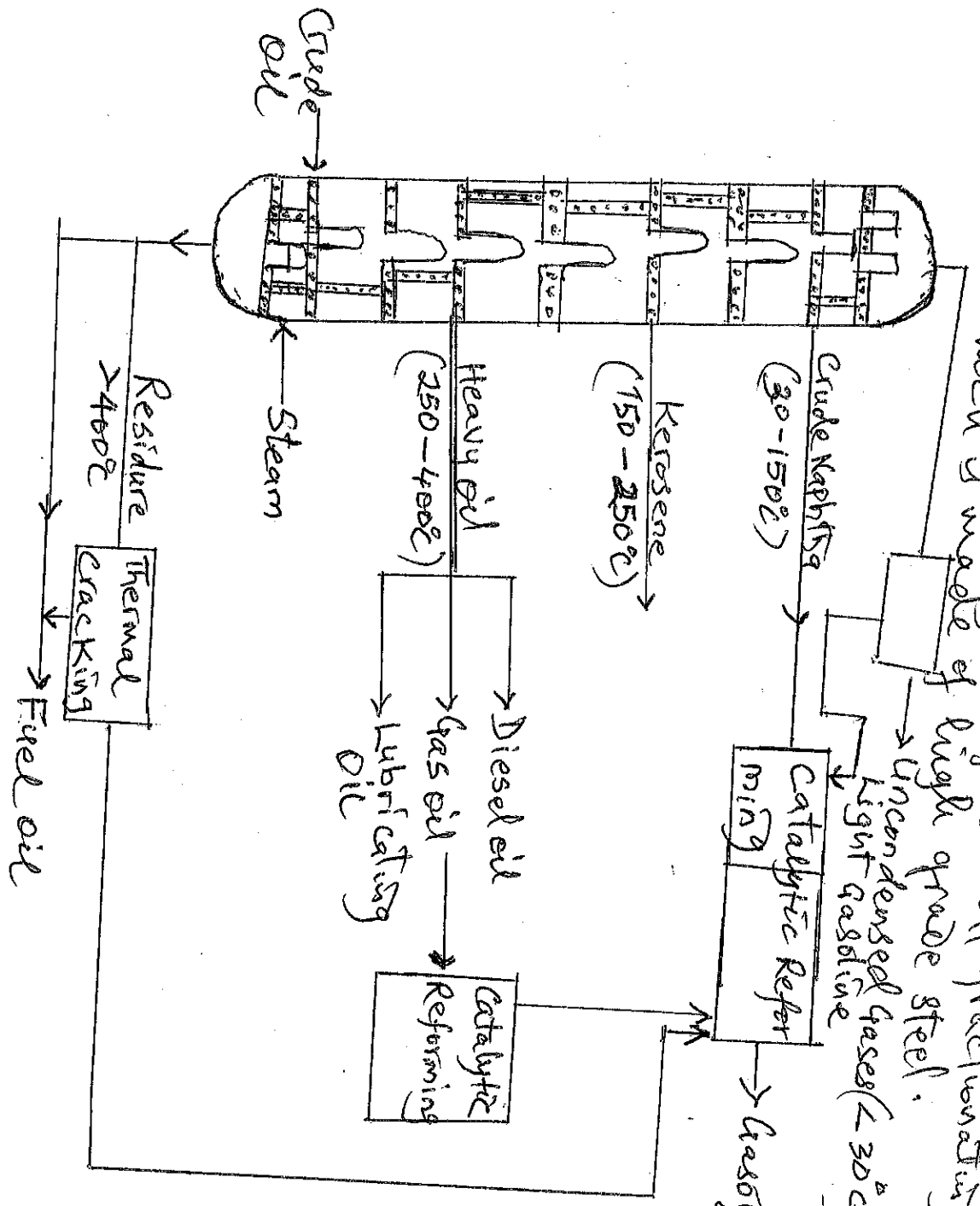
was dug by James M. Williams in 1857 in North America, and the oil was found at a depth of only 20m.

Refineries are mostly installed at a distance of few kilometers from the oil field for safety and fire hazard reasons. Transportation of petroleum through pipes does not involve too much expenditure, except for the initial investment for laying of the pipelines.

REFINING OF PETROLEUM

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The Petroleum obtained by mining is viscous and dark coloured. It has an unpleasant smell due to the presence of sulphur compounds, it also has impurities of sand and brine or sea water and is technically known as Crude oil. When, it is subjected to fractional distillation in a refinery, the Crude oil as it is practically useless, unless refined continuously in a specially designed tall fractionation tower column which is made of high grade steel.



Flow Sheet of Petroleum Refining

The crude oil is heated in a furnace to about 400°C and introduced into the tower. The tower is hot towards the lower end and comparatively cooler at the upper end. High boiling point fractions condense first and flow out at the lower end of the tower, while the low boiling fractions rise upwards and are collected as heavy condense. The ~~remaining~~ ^{fractions} of oil obtained above are also called 'straight run oil', and are not marketable without being further refined by physical or chemical methods. The uncondensed gases are liquefied to yield liquefied Petroleum gas (LPG) having mainly methane, ethane, Propane and butane and are supplied in cylinders under pressure for use mainly as domestic and industrial purposes.

Fractions	Boiling Range ($^{\circ}\text{C}$)	Approx. composition	Uses
1. Uncondensed gases (Cymene and Rhigolene)	$< 30^{\circ}\text{C}$	C_1 to C_5	fuel gases, Synthesis of organic chemicals, Refrigerants, Pigment carbon black.
2. Crude Naphtha Further fractionation gives: i) Petroleum ether ii) Petrol or Gasoline iii) Benzene	30-150 $^{\circ}\text{C}$ 120-150 $^{\circ}\text{C}$	$\text{C}_5 - \text{C}_{10}$ C_8 to C_{10}	Solvent for fat, oil, varnish, rubber fuel for car engines. Dry cleaning, solvent.
3. Kerosene Refractionation gives: i) Turbine fuel ii) Jet fuel	150-250 $^{\circ}\text{C}$ —	C_{10} to C_{15} —	Fuel, Illuminant, oil gas bag for Asbestos, cosmetics, oils liniments, etc. Aviation fuel after deodorisation.

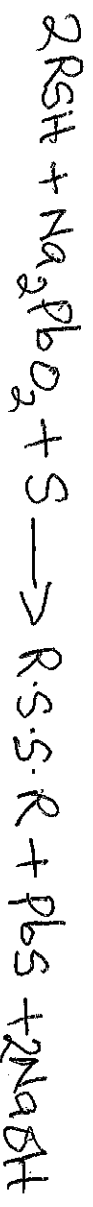
Fractions	Boiling Range	Approx. Composition	Uses
4. Heavy oil Refractionation gives: i) Gas oil ii) Fuel oil iii) Diesel oil	250-400°C	C ₁₂ -C ₃₁	fuel in diesel engine converted to gasoline by cracking.
5. Residual oil further fractionation gives: i) Lubricating oil ii) Paraffin wax iii) Vaseline iv) Pitch and Asphalt	Above 400°C	C ₁₇ -C ₄₀ C ₁₇ -C ₂₀ C ₂₀ -C ₃₀ C ₂₀ -C ₃₀ C ₃₀ -C ₄₀	Lubrication Candle industry, boot polish, wax paper. Paints, Trolleys, lubricants Road Surfacing

Other fractions may be having bad smell and dark coloured due to the presence of Sulphur containing compounds such as mercaptans and aromatic compounds respectively. These fractions are technically known as Sour and are purified for use.

PURIFICATION

1. Treatment with conc. Sulphuric acid: Here we give rise or remove aromatic compounds gets agitated with H₂SO₄ to sulphonated. It is then allowed to settle and the upper layer is withdrawn and treated with alkali to remove excess of acid; and finally washed with water.

ii) Sometimes sodium plumbate has to be added in the presence of alkali and calculated amount of sulphur to bring about the oxidation of mercaptans to disulphides in a process called (Docter Sweetening Process).



iii) Treatment with liquid sulphur dioxide: In the treatment of various fractions with sulphuric acid aromatic compounds are destroyed but when sulphur dioxide is used, the oil is purified without destroying the aromatic compounds.

Kerosene for e.g., is shaken with liquid sulphur dioxide and allowed to settle. The lower ^{layer} called 'extract' consisting of liquid sulphur dioxide with aromatic compounds, which

the upper layer called 'raffinate' is that of kerosene and rest of sulphur dioxide. When the two layers are separated, sulphur dioxide will be removed by heating each of the fractions. This process is called (Edelmann Process).

iv) Treatment with adsorbents: In order to remove persistent impurities and odours, various fractions are allowed to pass over adsorbents like Kieselguhr, clay, etc. and the resulting oils are generally purer.

INCREASING THE YIELD OF GASOLINE OR PETROL

Formally out of the various fractions of petroleum, kerosene oil was regarded as the most important but the advent of automobile in Britain gave the assumption of Petrol as the most important nowadays.

At present the demand for petrol is much greater than obtained by the fractionation of petroleum. It has therefore become necessary to find alternative methods of getting Petrol or to increase the production by converting some fractions of petroleum to petrol. The yield of petrol gets increased by converting higher boiling fractions like heavy oil to petrol and other synthetic methods.

Commonly employed methods for this process are: Cracking, Reforming and Isomerization.

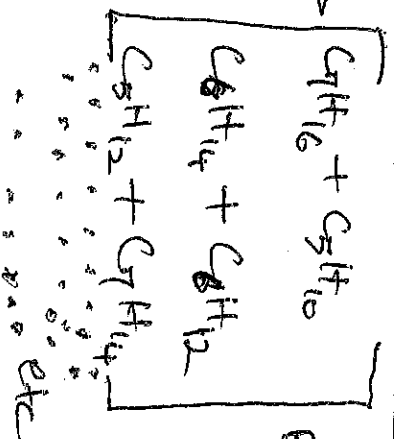
Cracking: This is the process of decomposition of less volatile higher hydrocarbons into more volatile lower hydrocarbons with the application of heat and catalyst.

As the yield of petrol from the distillation of petroleum is usually small (about 16%), more gasoline or petrol can be obtained by cracking less useful and cheap straight run oils like heavy oils and residual oils. In fact, approximately 50% of petrol is now obtained by this method. Cracking besides increasing the yield of petrol also tends to increase quality of it in terms of anti-knock properties. ~~The products~~

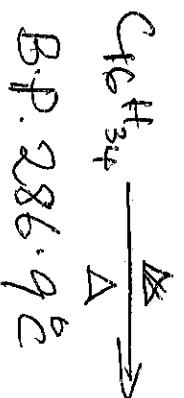
* The fractions like heavy oil and residual oil has ~~alter~~ alkanes or higher hydrocarbons having 12 or more carbon atoms. If they are subjected to cracking these large hydrocarbon molecules break up to yield hydrocarbons with lesser number of carbons. This decrease in size or number of carbons is accompanied with a change in the b.p range or volatility of the resulting product.

for example; $C_{12}H_{36}$ $\xrightarrow{\Delta}$

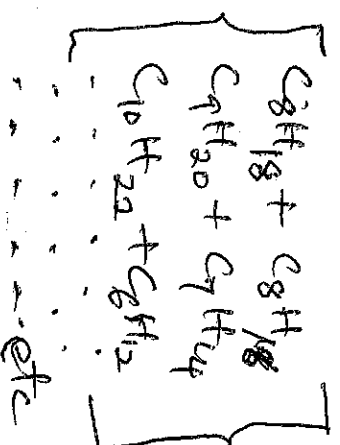
B.P. $216^{\circ}C$



Boiling range
30-150°C



B.P. $286.9^{\circ}C$



Boiling range
30-150°C

The nature of the product formed has been found to depend on the following:

- i) Structure of the hydrocarbon.
- ii) Temperature and pressure applied.
- iii) Presence or absence of catalyst.

Various atoms in a hydrocarbon molecule are held in their places by chemical bonds which are associated with certain amount of bond energy. When heated, the kinetic energy of the molecules increases as they are in constant motion and also having internal vibrations. If the kinetic energy is able to overcome the bond energy the molecules break up or the Carbon-Carbon bond gets ruptured to yield free radical fragments butanes and feed fragments of various valencies and hydrogen atoms. Sometimes stabilization gets accompanied with rearrangement and isomerization.